KATIXO KHOJLAB · NARRATION SCRIPT

EP06 — How does a computer store a photo?

ENGLISH

01 Title

How does a computer **store** a photo? You take a picture, it shows up in your gallery — but to the machine, that photo is just a grid of **numbers**. Today, let's zoom in and see them.

> **[CUE]** Hold on the title ~2s, calm curious tone, then go.

02 The Hook

Look at any photo. It looks smooth, real, full of colour. But here's the twist — a computer doesn't see a "picture" at all. It sees a giant **spreadsheet of numbers**. So let's zoom in, closer and closer, until you can actually see them.

> **[CUE]** Energy up — this is the hook. Mime a pinch-zoom on "zoom in".

03 Today's question

So here's today's question. Your selfie is colour and light. But a computer only understands **numbers**. So how on earth does it hold an entire photo? Let's find out.

> **[CUE]** Slow down. Let the question land before moving on.

04 ### Idea 1 — Pixels

Idea number one. Every image is a grid of tiny coloured squares, and each square is called a **`pixel`**. Zoom in far enough on any photo and you literally start seeing those squares. A single phone photo has **millions** of them. That's exactly what "**megapixels**" means — "mega" is just a fancy word for **million**. A twelve-megapixel camera captures twelve million pixels per shot.

> **[CUE]** Reveal the pixel chips one by one, then land on "megapixels".

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05 ### Idea 2 — RGB

Idea two. Each pixel is just one colour — and every colour is stored as **three numbers**: Red, Green, and Blue. Each one runs from **zero to 255**, from completely off to full brightness. Two-fifty-five, zero, zero is pure red. Zero, zero, zero — all lights off — is black. And all three at 255 is white. Mix those three numbers and you can make **any colour** on the screen.

> **[CUE]** Read the RGB triples off the code block; stress 0 and 255.

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06 ### Idea 3 — A big list of numbers

Now put it together. One photo is millions of pixels, and each pixel is three numbers — so the whole photo is one **enormous list of numbers**. And remember from earlier, every number is finally stored as **`binary`** — those same zeros and ones. That's why photos are "**heavy**". More pixels means more numbers means a bigger file — which is why we measure photos in **megabytes**.

> **[CUE]** Callback nod to the 0s-and-1s episode on "binary".

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07 Idea 4 — Compression

Idea four. Storing every single number is wasteful — so we get clever with **compression**. A format like **JPEG** does two smart things: it throws away tiny details your eye would never miss, and it reuses repeated patterns instead of writing them out again and again. That's how a JPEG ends up **far smaller** than the raw image. The trade-off: **PNG** keeps **everything** — that's lossless. JPEG drops a little detail to save space — that's lossy.

> **[CUE]** Reveal the PNG-vs-JPEG note card on "lossless / lossy".

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08 Why it matters

And suddenly your whole gallery makes sense. **Megapixels** tell you how many pixels the camera grabs. **File size** tells you how many numbers had to be stored. Open an old, tiny photo or zoom in way too far, and you see the squares — that "**blocky**" look is just pixels getting big. And when you share a logo, pick PNG so it stays crisp; for a normal photo, JPEG saves you space.

> **[CUE]** Point to each bullet as you hit megapixels / size / blocky / PNG.

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09 Common myth

Which brings us to the myth. People think the computer stores the actual colours — that the real picture is somehow sitting inside the file. It isn't. It stores **numbers** — the red, green, blue values for every pixel. Your **screen** is what turns those numbers back into light. The "photo" only really exists the moment it's **displayed**.

> **[CUE]** Beat on the myth, then reveal the green "Reality" card.

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10 Recap

Quick recap. One: a photo is a grid of **pixels**, and each pixel is just a few numbers — its red, green, and blue. Two: **compression** shrinks the file by cleverly dropping detail you won't even notice.

> **[CUE]** Crisp and clear — these are the two lines to remember.

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11 CTA

And that’s it — your photo is really just a giant grid of numbers, and now you can see them. If this changed how you look at a picture, subscribe to Katixo KhojLab. Next question: what is ”the cloud” — actually? See you there.

> **[CUE]** Warm, friendly. Smile in your voice on the subscribe line.
